An architecture-independent dissipation floor for alternating-renewal observations

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Abstract

Can hidden Markov architectures of increasing size reproduce a fixed observed process with arbitrarily small dissipation? For finite-realizable stationary two-label alternating-renewal laws, we prove that zero all-dimension cost is equivalent to both complete dwell laws being finite positive mixtures of exponential densities, and to the existence of a finite reversible exact realization. The necessity extends the established equilibrium spectral restriction to vanishing-dissipation sequences of unbounded dimension and unrestricted rates. Comparing a chain with its additive reversibilization converts negative correlation between bounded past and future persistence tests into a dimension-independent dissipation certificate. For a specified three-state-ring observation at $q = 1/20$, an exact-arithmetic fixed-window certificate gives $V_\infty > 0.017568198744$ in target time units, with Boltzmann's constant one. We also give an example beyond the elementary variance obstruction and a certificate stable under bounded observational errors. The exact positive optimum, its finite-dimensional attainment, and statistical confidence-set construction remain unresolved.

1 The question and its resolution

Let Q be a finite irreducible continuous-time Markov generator in row convention, $\pi Q = 0$, and let h assign a visible label to every microscopic state. The chain starts in its stationary law. Write

$$F_{ij} = \pi_i Q_{ij}, \qquad \sigma(Q) = \sum_{i<j} (F_{ij} - F_{ji}) \log \frac{F_{ij}}{F_{ji}}. \tag{1}$$

A missing reverse edge carrying positive forward flux gives infinite cost. Zero flux in both directions gives zero contribution. All logarithms are natural. Ordinary time reversal is used; microscopic variables are even under time reversal.

For a stationary observed path law P, define

$$V_n(P) = \inf\{\sigma(Q) : |Q| \le n,\ P_{Q,h} = P\}, \qquad V_\infty(P) = \inf_n V_n(P). \tag{2}$$

Equality $P_{Q,h} = P$ means equality of the entire stationary observed path law, not merely a collection of moments. The target and these conventions are those of Aznagulov [1]. Its unpriced zero-limit question is resolved here without using its fixed-dimension compactification claims.

Theorem 1 (Main result)**.** *For the alternating-renewal law P_q defined in Section 2, and every $0 < q < 1$,*

$$0 < \frac{2(1-q)^4}{3(3q+5)^2(2q^2+3q+3)^2} \le V_\infty(P_q) \le (1-q)\log(1/q). \tag{3}$$

At $q = 1/20$ the following stronger rational certificate holds:

$$V_\infty(P_{1/20}) \geq \frac{182774933998705388800}{22434692075028399713067} > \frac{1}{125}. \tag{4}$$

Every inequality is valid over the unrestricted finite deterministic-observation class, including realizations with multiple microscopic states under either visible label.

The lower bounds are not asserted to be sharp. In particular, this theorem decides whether the all-dimension infimum is zero, but does not distinguish a positive minimum attained at finite dimension from a positive infimum approached only at growing dimension.

The central structural result is Theorem 9: for every finite-realizable stationary ergodic two-label alternating-renewal law,

$$\begin{aligned} V_\infty(P) = 0 \quad &\iff \quad \text{both dwell laws are hyperexponential} \\ &\iff \quad \text{a finite reversible exact realization exists.} \end{aligned}$$

Here hyperexponential means a finite positive mixture of exponential densities, not an arbitrary phase-type law. The equilibrium mixture restriction is classical; the extra implication is that a sequence of exact realizations with vanishing dissipation must enter this class even without a common bound on dimension or rates. Section 7.1 illustrates a case not detected by the elementary dwell-variance test.

2 The target law and an exact finite competitor

For $q > 0$ put

$$s = 1 + q, \qquad c = 1 + q + q^2, \qquad D_q(z) = z^2 + 2sz + c.$$

The complete A visits are exponential with rate s. The complete B visits have Laplace transform

$$\widehat{f}_q(z) = \frac{(2q/s)z + c}{D_q(z)}. \tag{5}$$

Successive visits are independent and alternate between A and B. The stationary version of this renewal process defines P_q.

These specifications have the exact three-state realization

$$Q_q = \begin{pmatrix} -s & 1 & q \\ q & -s & 1 \\ 1 & q & -s \end{pmatrix}, \qquad \pi = (1/3, 1/3, 1/3), \qquad h(0) = A, \quad h(1) = h(2) = B. \tag{6}$$

For a complete B visit the entrance distribution, killed generator, and exit rates are

$$\alpha = (1/s, q/s), \qquad T = \begin{pmatrix} -s & 1 \\ q & -s \end{pmatrix}, \qquad t = (q, 1)^{\mathsf{T}}.$$

Direct inversion gives $\alpha(zI - T)^{-1}t = \widehat{f}_q(z)$. The single A state resets the process on every return; the holding time in A is independent of its exit destination. Thus the construction realizes the complete alternating-renewal law, not just its marginal waiting-time distributions.

Differentiating (5) gives

$$\mathbb{E}T_A = 1/s, \qquad \mathbb{E}T_B = 2/s, \qquad \mathbb{E}T_B^2 = 6/c. \tag{7}$$

The stationary B probability is $p = 2/3$ and the rate of entrances into each block is $r = s/3$. Each of the three unoriented edges in (6) carries opposing fluxes $1/3$ and $q/3$. Consequently

$$\sigma(Q_q) = (1 - q)\log(1/q), \tag{8}$$

proving the upper bound in (3). In particular, the all-dimension infimum is finite. At $q = 1$ the B dwell reduces to an exponential of rate one; a reversible two-state chain realizes the target.

3 Small entropy production forces proximity to hidden equilibrium

For an arbitrary realization Q, define its stationary reverse and additive reversibilization by

$$Q^\dagger = D_\pi^{-1} Q^\mathsf{T} D_\pi, \qquad R = \tfrac{1}{2}(Q + Q^\dagger). \tag{9}$$

The generator R is irreducible and reversible for π. It has the same diagonal entries as Q and therefore the same exit rate from every microscopic state. It is a comparison chain, not an exact realization of the target.

Lemma 2 (Dimension-free path comparison)**.** *Let $\mathcal{P}_Q^T$ and $\mathcal{P}_R^T$ be the stationary microscopic path laws on a time interval of length T. Then*

$$D_{\mathrm{KL}}(\mathcal{P}_Q^T \| \mathcal{P}_R^T) \le \frac{T}{4}\sigma(Q). \tag{10}$$

If two independent exponential clocks of rate $z > 0$ delimit the interval $[-E_-, E_+]$, independently of the chain, the corresponding joint laws of clocks and path satisfy

$$D_{\mathrm{KL}}(\mathcal{P}_Q^{\mathrm{clock}} \| \mathcal{P}_R^{\mathrm{clock}}) \le \frac{\sigma(Q)}{2z}. \tag{11}$$

Proof. The initial stationary laws coincide, and the holding-time terms cancel because the exit rates agree. The standard jump-path likelihood ratio, or its direct product-of-rates calculation, gives

$$\frac{1}{T} D_{\mathrm{KL}}(\mathcal{P}_Q^T \| \mathcal{P}_R^T) = \sum_{i \ne j} F_{ij} \log \frac{2F_{ij}}{F_{ij} + F_{ji}}.$$

For $x, y > 0$, the scalar inequality

$$x \log \frac{2x}{x+y} + y \log \frac{2y}{x+y} \le \frac{1}{4}(x - y)\log(x/y) \tag{12}$$

suffices. To verify it, set $m = (x + y)/2$ and $u = (x - y)/(x + y)$. The right side minus the left, divided by m, is

$$L(u) = -\log(1 - u^2) - u \operatorname{artanh} u.$$

This function is even and $L(0) = 0$. For $0 \le u < 1$,

$$L'(u) = \frac{u}{1 - u^2} - \operatorname{artanh} u \ge 0,$$

since the expression on the right vanishes at zero and has derivative $2u^2/(1 - u^2)^2$. This proves (12). If $\sigma(Q) = \infty$, the claimed bound is immediate; otherwise all used edges have reverse support, and summing (12) proves (10).

Condition on the independently drawn clock lengths and use stationarity to translate the interval to one beginning at zero. The mean total length is $2/z$. Integrating (10) proves (11). No optional-stopping identity or assumption about independent path increments is being used. □

The factor 1/4 is a rate-level comparison with the generator R. It is not a statement that the path law of R is the arithmetic mixture of the forward and reversed path laws.

4 Two observable clocks and an equilibrium restriction

Let B be any nontrivial observed block, with stationary probability p. Define its stationary persistence function

$$H(t) = \mathbb{P}\{Y_u = B \text{ for every } 0 \le u \le t\}, \qquad H(0) = p. \tag{13}$$

Independently draw $E_-, E_+ \sim \mathrm{Exp}(z)$. Conditional on $Y_0 = B$, define the binary tests

$$U = \mathbf{1}\{Y_u = B \text{ throughout } [-E_-, 0]\},$$
$$V = \mathbf{1}\{Y_u = B \text{ throughout } [0, E_+]\}.$$

The tests concern an uninterrupted visit. Seeing B at the two endpoints alone is not sufficient. Hidden jumps within B are allowed. The clocks are an external mathematical sampling device, not a modification of the physical realization.

Stationarity gives equal marginals. Writing $\widehat{H}(z) = \int_0^\infty e^{-zt} H(t)\, dt$, define

$$a = \mathbb{P}(U = 1 \mid B) = \mathbb{P}(V = 1 \mid B) = \frac{z\widehat{H}(z)}{p}, \qquad b = \mathbb{P}(U = V = 1 \mid B) = -\frac{z^2 \widehat{H}'(z)}{p}. \tag{14}$$

For the joint event, stationarity gives persistence over an interval of length $E_- + E_+$, whose density is $z^2 t e^{-zt}$. Thus (14) follows directly, without a renewal assumption.

Lemma 3 (Equilibrium has nonnegative clock covariance)**.** *For every finite reversible hidden chain with deterministic observation, the two-clock probabilities satisfy $b \ge a^2$.*

Proof. Conditional on a present microscopic state $X_0 = i$, the stationary past and future are independent. Reversibility makes their conditional laws identical. After including the independent clocks, both binary tests have the same conditional success probability $v_i \in [0, 1]$. Hence, conditional on $Y_0 = B$,

$$a = \sum_{i \in B} \frac{\pi_i}{p} v_i, \qquad b = \sum_{i \in B} \frac{\pi_i}{p} v_i^2,$$

and the assertion is the nonnegativity of the variance of v_i. □

5 A quantitative information-projection lemma

Suppose $0 < a < 1$, $b < a^2$, and set $\delta = a^2 - b > 0$. The four conditional probabilities, ordered as $00, 01, 10, 11$, are

$$w = \big((1-a)^2 - \delta,\ a(1-a) + \delta,\ a(1-a) + \delta,\ a^2 - \delta\big). \tag{15}$$

Assume they are positive; boundary cases follow by limits when needed. Let

$$r_a = \big((1-a)^2,\ a(1-a),\ a(1-a),\ a^2\big), \qquad I(a, b) = D_{\mathrm{KL}}(w \| r_a).$$

The quantity $I(a, b)$ is the mutual information of the two binary tests.

Lemma 4 (Nearest positively correlated symmetric binary law)**.** *For any valid symmetric binary law*

$$r_{u,v} = (1 - 2u + v, u - v, u - v, v), \qquad v \geq u^2,$$

one has

$$D_{\mathrm{KL}}(w\|r_{u,v}) \geq I(a, b). \tag{16}$$

Thus the closest law in this class, in the stated direction of relative entropy, is the independent law with the target marginal a.

Proof. If a zero coordinate of $r_{u,v}$ has positive w mass, the claim is immediate. Otherwise $-\log x \geq 1 - x$ gives

$$\begin{aligned} D_{\mathrm{KL}}(w\|r_{u,v}) - D_{\mathrm{KL}}(w\|r_a) &\geq 1 - \sum_j w_j \frac{(r_{u,v})_j}{(r_a)_j} \\ &= \frac{\delta[(v - u^2) + (u - a)^2]}{a^2(1 - a)^2} \geq 0. \end{aligned}$$

Equality is attained at $u = a, v = a^2$. □

Lemma 5 (An exact rational lower bound for the information)**.** *For $\delta = a^2 - b > 0$ as above,*

$$I(a, b) \geq \frac{\delta^2}{2a^2(1 - a)^2} + \frac{(2a - 1)^2\delta^3}{6a^4(1 - a)^4}. \tag{17}$$

In particular $I(a, b) \geq 8\delta^2$.

Proof. Keep a fixed and replace δ in (15) by $x \in [0, \delta]$. Let the resulting information be $I(x)$. Then $I(0) = I'(0) = 0$, and

$$I''(x) = \frac{1}{(1 - a)^2 - x} + \frac{2}{a(1 - a) + x} + \frac{1}{a^2 - x}.$$

Use, on their respective domains,

$$\frac{1}{r - x} \geq \frac{1}{r} + \frac{x}{r^2}, \qquad \frac{1}{r + x} \geq \frac{1}{r} - \frac{x}{r^2}.$$

It follows that

$$I''(x) \geq C_0 + C_1 x, \quad C_0 = \frac{1}{a^2(1 - a)^2}, \quad C_1 = \frac{(2a - 1)^2}{a^4(1 - a)^4}.$$

Twice integrating proves (17). Finally $a(1 - a) \leq 1/4$ gives $C_0/2 \geq 8$. □

Theorem 6 (Dimension-free observable dissipation certificate)**.** *For any finite stationary hidden Markov realization and the observed two-clock statistics above, if $b < a^2$ then*

$$\sigma(Q) \geq 2zp\, I(a, b), \tag{18}$$

$$\sigma(Q) \geq zp\frac{\delta^2}{a^2(1 - a)^2} + \frac{zp}{3}\frac{(2a - 1)^2\delta^3}{a^4(1 - a)^4}. \tag{19}$$

The bound requires no ceiling on hidden dimension, rates, or stationary activity.

Proof. Compare Q with its reversibilization R on the random two-sided interval. Both chains have the same stationary mass p in B. Apply data processing to the record consisting of the event $Y_0 \in B$ and, on that event, the two tests. The chain rule for relative entropy and Lemma 2 give

$$\frac{\sigma(Q)}{2z} \geq p\, D_{\mathrm{KL}}(w \| w_R).$$

Under R, Lemma 3 gives the required symmetric positively correlated law w_R. Lemma 4 proves (18), and Lemma 5 proves (19). All statements hold for an arbitrary original realization; R need not reproduce the target law. □

6 Evaluation for the full family and the specified parameter

For an alternating-renewal process, a stationary time falls uniformly along a length-biased visit. Equivalently, the expected fraction of starting times for which an interval of length t lies in a complete B visit of duration u is proportional to $(u-t)_+$. Therefore

$$H_q(t) = r \int_t^\infty (u-t) f_q(u)\, du, \qquad r = s/3.$$

Using (5) gives the particularly simple transform

$$\widehat{H}_q(z) = \frac{p}{z} - \frac{r(1-\widehat{f}_q(z))}{z^2} = \frac{2z+3s}{3D_q(z)}. \tag{20}$$

Substitution in (14) yields

$$a_q(z) = \frac{z(2z+3s)}{2D_q(z)}, \tag{21}$$

$$b_q(z) = \frac{z^2(z+q+2)(z+2q+1)}{D_q(z)^2}, \tag{22}$$

$$\delta_q(z) = a_q(z)^2 - b_q(z) = \frac{z^2(1-q)^2}{4D_q(z)^2} > 0 \quad (q \neq 1). \tag{23}$$

Thus every positive clock rate supplies an observable equilibrium obstruction.

Keeping only the quadratic term of (19) gives, for every $z > 0$,

$$V_\infty(P_q) \geq \frac{2z^3(1-q)^4}{3(2z+3s)^2(sz+2c)^2}. \tag{24}$$

Setting $z = 1$ proves the lower bound in (3).

For the concrete target, choose the fixed clock rate $z = 6$. Exact arithmetic gives

$$p = \frac{2}{3}, \qquad a = \frac{18180}{19861}, \qquad b = \frac{329212800}{394459321}, \qquad \delta = \frac{1299600}{394459321}. \tag{25}$$

The four joint probabilities and their independent comparison probabilities share the denominator $394459321 = 19861^2$:

	00	01	10	11
Target numerator	1526161	31860180	31860180	329212800
Independent numerator	2825761	30560580	30560580	330512400

Equation (19) is now a calculation involving rational numbers only. It gives

$$\sigma(Q) \geq \frac{182774933998705388800}{22434692075028399713067} = 0.008146977608939347\ldots.$$

The strict margin over $1/125$ is exactly

$$\frac{412174674809773886933}{28043365093785499641333375} > 0.$$

This proves (4) and finishes Theorem 1.

Retaining the logarithms in (18), rather than the rational underestimate, gives the stronger explicit certificate

$$V_\infty(P_{1/20}) \geq \frac{8}{394459321}\Bigg[1526161\log\frac{1526161}{2825761} + 63720360\log\frac{31860180}{30560580} + 329212800\log\frac{329212800}{330512400}\Bigg] = 0.0084471687960995336\ldots.$$

A rational series with an explicit remainder certifies this numerical enclosure in the supplied verifier. This is a certified lower bound, not the evaluated optimum.

6.1 A stronger fixed-window certificate

The exponential clocks are not essential. For any $t > 0$, use persistence on $[-t, 0]$ and $[0, t]$, conditional on $Y_0 = B$, and set $a_t = H(t)/p$, $b_t = H(2t)/p$.

Corollary 7 (Fixed-window bound)**.** *If $b_t < a_t^2$, then*

$$\sigma(Q) \geq \frac{2p}{t} I(a_t, b_t).$$

The same finite stationary deterministic-observation assumptions apply; renewal is not required.

Proof. The comparison path interval has length $2t$. Lemma 2 and conditional data processing give $pD_{\mathrm{KL}}(w\|w_R) \leq t\sigma(Q)/2$. Given the current microscopic state, reversible past and future have the same independent persistence law, so Lemmas 3 and 4 apply without change. □

For the ring target, $g(t) = H_q(t)/p = \frac{1}{2}\mathbf{1}^{\mathsf{T}} e^{tT}\mathbf{1}$. At $q = 1/20$ and $t = 3/10$, exact rational enclosures certify

$$0.017568198744 < B_{3/10} := \frac{2p}{t} I(g(t), g(2t)) < 0.017568198745.$$

Thus $V_\infty(P_{1/20}) > 0.017568198744$, improving the displayed exponential-clock certificate by a factor of about 2.080. This extension was suggested by the supplied referee audit and incorporated in version 1.1.

For reproducibility, put $x = st$ and $C = I + T/s$, which is substochastic. Uniformisation gives $g(t) = e^{-x}\sum_{k\geq 0}(x^k/k!)c_k$, with $c_k = \frac{1}{2}\mathbf{1}^{\mathsf{T}} C^k \mathbf{1} \in [0, 1]$. For both required times, $x \leq 1$. Consecutive alternating Taylor sums enclose e^{-x}; truncating the positive series at $N = 32$ leaves a tail at most

$$\frac{x^{N+1}}{(N+1)!}\left(1 - \frac{x}{N+2}\right)^{-1}.$$

For rational $r > 0$, $y = (r-1)/(r+1)$ gives $\log r = 2\sum_{k=0}^{m-1} y^{2k+1}/(2k+1) + R_m$, with $|R_m| \leq 2|y|^{2m+1}/((2m+1)(1-y^2))$. Rational interval operations certify the enclosure, including

negative logarithmic terms (the supplied audit uses $m = 32$ and the release implementation uses $m = 40$). The upper enclosure is for this certificate, not for V_∞. The physical bracket is instead

$$0.017568198744 < V_\infty(P_{1/20}) \le (19/20)\log 20 \approx 2.84595.$$

An explicit certificate, the best certificate in a selected observation family, and the unknown physical optimum are different quantities. The gap to the ring competitor is not an error estimate for the lower certificate.

6.2 Bounded observational uncertainty

Corollary 8 (A conservative uncertainty-box certificate)**.** *Suppose the true stationary statistics obey $p \ge p_0 > 0$, $a \ge a_0 \ge 0$, $b \le b_1$, and $\eta = a_0^2 - b_1 > 0$. Then exponential clocks of rate z give $\sigma(Q) \ge 16 z p_0 \eta^2$. For fixed windows of length t on each side, $\sigma(Q) \ge 16 p_0 \eta^2 / t$.*

Proof. Every valid law in the box has $a^2 - b \ge \eta$. Use $I(a, b) \ge 8(a^2 - b)^2$ in Theorem 6 or Corollary 7; boundary cells follow by continuity. □

For the $q = 1/20, z = 6$ witness, allow absolute errors at most $\varepsilon = 1/10000$ in each of p, a, b. The choices $p_0 = 2/3 - \varepsilon$, $a_0 = 18180/19861 - \varepsilon$, $b_1 = 329212800/394459321 + \varepsilon$ give $\eta > 0$ and a strictly positive rational floor, verified by the extension checker. This is a deterministic sensitivity statement. A confidence claim additionally needs simultaneous coverage for this box, justified for the observation scheme and temporal dependence; no such statistical procedure or sample-size guarantee is asserted here. A box with $\eta \le 0$ supplies no positive certificate by this test, not evidence of equilibrium.

7 A broader zero-cost dichotomy

The positive floor is not restricted to this example. A complete dwell distribution is called *hyperexponential* when its density is a finite positive mixture of exponential densities, $\sum_j \alpha_j \lambda_j e^{-\lambda_j t}$, with $\alpha_j > 0$, $\sum_j \alpha_j = 1$, and $\lambda_j > 0$.

Theorem 9 (No unattained zero limit for finite-realizable alternating renewal)**.** *Let P be a stationary ergodic two-label alternating-renewal law that has a finite irreducible CTMC realization with deterministic observation. The following are equivalent:*

(i) $V_\infty(P) = 0$*;*

(ii) both complete dwell distributions are hyperexponential;

(iii) P has a finite reversible exact realization.

Thus growing architecture cannot approach zero entropy production without a finite zero-cost optimizer anywhere in this class.

Proof. The implication (iii)⇒(i) is immediate. To prove (i)⇒(ii), choose exact realizations Q_k with $\sigma(Q_k) \to 0$, and replace each by $R_k = (Q_k + Q_k^\dagger)/2$. For either observed block B, the stationary mass is the same fixed p. By (10) and Pinsker's inequality,

$$|H_{R_k}(t) - H_P(t)| \le \sqrt{t\sigma(Q_k)/8}. \tag{26}$$

The killed generator $(R_k)_{BB}$ is self-adjoint in $L^2(\pi|_B)$ and strictly negative definite. Here $\pi|_B$ is the unnormalised restricted measure, of mass p, not the conditional probability measure. Its spectral theorem gives

$$H_{R_k}(t) = \langle \mathbf{1}, e^{t(R_k)_{BB}}\mathbf{1}\rangle_{\pi|_B} = \int_{[0,\infty)} e^{-\lambda t}\,\mu_k(d\lambda), \qquad \mu_k([0,\infty)) = p,$$

with a finite positive discrete measure μ_k.

For completeness, the limit retains a positive spectral measure. Compactify $\lambda \in [0,\infty)$ by the coordinate $x = e^{-\lambda} \in (0,1]$. A subsequence of the corresponding measures on $[0,1]$ converges weakly. For each $t > 0$, x^t is continuous, so (26) identifies $H_P(t)$ with its limiting integral. Continuity $H_P(t) \to p$ as $t \downarrow 0$ excludes mass at $x = 0$. Also $H_P(t) \to 0$ as $t \to \infty$ excludes mass at $x = 1$. Hence

$$H_P(t) = \int_{(0,\infty)} e^{-\lambda t}\,\mu(d\lambda)$$

for a positive finite measure of mass p.

Now use finite realizability of the fixed target, not of the approximating reversible sequence. For any one finite target realization, $H_P(t) = \pi_B e^{tQ_{BB}}\mathbf{1}$, so H_P satisfies a nonzero constant-coefficient differential equation $P_0(\partial_t)H_P = 0$, by Cayley–Hamilton. Differentiating the integral at $t > 0$ is legitimate because polynomial factors times $e^{-\lambda t}$ are bounded. Thus

$$\int P_0(-\lambda)e^{-\lambda t}\,\mu(d\lambda) = 0 \qquad (t > 0).$$

The finite signed measure $e^{-\lambda}P_0(-\lambda)\mu(d\lambda)$ has zero Laplace transform. Uniqueness of that transform implies it is zero. Therefore μ is supported on the finitely many positive roots of $P_0(-\lambda)$, and H_P is a finite positive mixture of exponentials.

For a stationary complete dwell density f_B and entrance intensity r_B, the visit-length identity is $H_P''(t) = r_B f_B(t)$. If $H_P(t) = \sum_j c_j e^{-\lambda_j t}$, the density weights are $\alpha_j = c_j\lambda_j/r_B > 0$: indeed $f_B(t) = \sum_j \alpha_j\lambda_j e^{-\lambda_j t}$ and $\sum_j \alpha_j = -H_P'(0)/r_B = 1$. Apply the same argument to A to prove (ii).

Finally suppose (ii) holds, with densities $\sum_i \alpha_i\lambda_i e^{-\lambda_i t}$ in A and $\sum_j \beta_j\mu_j e^{-\mu_j t}$ in B. Introduce states A_i, B_j and rates

$$Q_{A_iB_j} = \lambda_i\beta_j, \qquad Q_{B_jA_i} = \mu_j\alpha_i,$$

with no within-block jumps. Set

$$r = \left(\sum_i \alpha_i/\lambda_i + \sum_j \beta_j/\mu_j\right)^{-1}, \quad \pi_{A_i} = r\alpha_i/\lambda_i, \quad \pi_{B_j} = r\beta_j/\mu_j.$$

Every cross-edge has flux $r\alpha_i\beta_j$ in both directions. The chain is irreducible and reversible. Each exit chooses the next block's phase independently from its specified mixture weights. This gives the exact independent alternating-renewal law, proving (iii). □

The renewal hypothesis is used for the sufficiency construction. For a general observed process, hyperexponential marginal dwell laws alone do not determine the complete path law and are not asserted to guarantee a reversible realization.

7.1 Beyond the elementary variance obstruction

Let the A dwell be exponential of rate one and let the B dwell be the equal mixture of the $q = 1/4$ ring dwell and an exponential of rate $1/100$. Its density is

$$f_B(t) = \frac{81}{160}e^{-3t/4} - \frac{49}{160}e^{-7t/4} + \frac{1}{200}e^{-t/100}.$$

An exact finite irreducible realization, with the first state labelled A and the other three labelled B, is

$$Q = \begin{pmatrix} -1 & 2/5 & 1/10 & 1/2 \\ 1/4 & -5/4 & 1 & 0 \\ 1 & 1/4 & -5/4 & 0 \\ 1/100 & 0 & 0 & -1/100 \end{pmatrix}.$$

The unique A state resets independent visits. All used edges have reverse support, so its dissipation is finite. Direct integration gives $\mathbb{E}T_B = 254/5$ and $\mathbb{E}T_B^2 = 70016/7$, hence

$$\mathrm{CV}_B^2 = \frac{324697}{112903} > 1.$$

The elementary necessary equilibrium condition $\mathrm{CV}^2 \geq 1$ therefore does not exclude this law. Nevertheless its Laplace transform has the negative residue $-49/160$ at $-7/4$. Uniqueness of the Laplace transform excludes any positive exponential-mixture representation. Theorem 9 implies $0 < V_\infty(P) < \infty$. This example demonstrates the classification's reach; it does not assign a numerical optimum or assert that the two-test certificate detects every nonhyperexponential law.

8 Interpretation, verification, and claim limits

A stationary alternating-renewal observation can be time-reversal symmetric even when every compatible Markov mechanism dissipates. The obstruction here is not an observed directional current. It is a restriction imposed by conditional independence and reversibility of the microscopic mechanism. Mixing equilibrium hidden states produces nonnegative past–future correlation for the two persistence tests, whereas P_q has the strictly negative covariance (23). The rate comparison (11) converts that separation into a uniform thermodynamic cost.

The inherited and additional implications should be distinguished. Tu [4] connects nonequilibrium switching to dwell-time restrictions. Skinner and Dunkel [2], Supplement Section II, equations S6–S10 (A6–A10 in the HTML rendering), derive the equilibrium positive exponential-mixture restriction and credit an earlier antecedent. Their waiting-time programme includes hidden-model optimization and a variance–dissipation curve. The $q = 1/20$ target already has $\mathrm{CV}_B^2 = 481/842 < 1$, so recognizing it as nonequilibrium is not the new conclusion. Ehrich [3] studies fitting hidden mechanisms for discrete-time partial observations and discusses enlarging the hidden network. Nitzan, Ghosal, and Bisker [5] optimize transition and waiting-time statistics with a hypothesized hidden topology. Neither numerical convergence in explored dimensions nor fixed-dimension exclusion is used here as a proof of a positive unrestricted infimum.

Statement	Observation/domain	Status here
Equilibrium spectral restriction	Dwell laws of a finite reversible chain	Inherited, explicitly credited
Hidden-model optimization	Statistics matched within explored hidden mechanisms	Prior methodology, not a new invention
Dimension-free certificate	Bounded persistence statistics; any finite hidden dimension	Explicit analytic separation and certified constants
Zero-cost closure	Exact finite-realizable alternating-renewal path law; unbounded approximating dimension	Vanishing cost forces the spectral class and a finite reversible realization

Additive reversibilization also appears in Aznagulov's finite-resolution comparison [6]; its use is not itself claimed new. The present argument addresses the zero-limit possibility of [1] without relying on that preprint's compactification claims. The source comparison is bounded, not an exhaustive priority certification or a claim that our certificate dominates every earlier estimator.

The evidence directory contains four complementary checks. A standard-library verifier evaluates the explicit certificate with exact fractions, including its strict margin above $1/125$ and a rigorous logarithmic enclosure. A second exact checker encloses the fixed-window certificate, verifies the high-variance example and the uncertainty-box substitution, and rejects six invalid inputs or false claims. A SymPy verifier checks thirteen all-parameter rational identities. A separately implemented floating-point diagnostic tests 180 random hidden chains, 1000 information-projection instances, and 32 ring-family cases, with explicit controls for one-way edges and equilibrium. These calculations do not replace the analytic proofs; they check algebra, constants, and failure-prone implementation details. No claim of external independent review or proof-assistant formalization is made.

The precise resolved claim is $V_\infty(P_{1/20}) \neq 0$, with a positive dimension-independent lower bound. The exact value of V_∞, whether a finite architecture attains this positive value, and the best hidden architecture remain distinct optimization questions. Nothing in the argument limits physical models with odd time-reversal variables, nonstationary driving, or observation rules outside the stated model class.

Availability and disclosure. Source, exact checks, diagnostics and a point-by-point revision record accompany this anonymous unrefereed candidate. The supplied review and audit informed revision; AI-assisted revision and internal checking are not external peer review. The stronger fixed-window corollary is credited to that audit. No human research contribution, external funding, institutional affiliation or conflict-of-interest declaration is inferred from the publisher account; those details were not supplied. This theoretical study involves no human participants or private empirical data. Original prose and data are CC0; original code is MIT; third-party material retains its terms.

References

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